



# Jarn de Jong

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Quantum Physicist,  
Cryptographer, Developer



[List of publications](#)

## Bio

Quantum physicist with a strong background in a wide range of topics. Software developer & tech enthusiast with a strong track record in Python, self-hosting, Docker and more.

Phd (summa cum laude) in quantum communication & cryptography, BSc. & MSc. in Applied Physics.

## Programming & Computer skills

Python



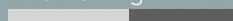
Linux/bash



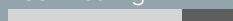
Docker



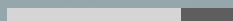
Networking



Self-hosting



Latex



## Quantum skills

Networking & communication  
Stabilizer & graph states  
Multipartite entanglement  
Local Equivalence  
QKD & CKA  
Anonymity in quantum networks  
Photonic systems  
Distributed systems  
QECC & Fault-tolerance  
State- and Process tomography  
Quantum Internet use cases

## Languages

**Dutch** Native  
**English** C2  
**German** B1-B2

## Experiences

|                                                                                                  |  |                               |                                         |
|--------------------------------------------------------------------------------------------------|--|-------------------------------|-----------------------------------------|
| 2022-2025                                                                                        |  | Quantum Internet Alliance     | Use Case team member                    |
| Project management, use case development and simulation, reporting.                              |  |                               |                                         |
| May 2023                                                                                         |  | LIP6 (Sorbonne Université)    | Visiting researcher                     |
| Research visit at the LIP6 team. Present earlier work and collaborate with multiple researchers. |  |                               |                                         |
| 2020-2025                                                                                        |  | Technische Universität Berlin | Scientific Employee                     |
| Designing & giving lectures, student supervision, scientific talks.                              |  |                               |                                         |
| 2019-2020                                                                                        |  | QuTech                        | Research assistant & software developer |
| Hardware- and software development, simulations. Work span off into Q*Bird.                      |  |                               |                                         |
| 2019                                                                                             |  | TNO                           | Quantum Technology Intern               |
| Design and develop experiments for upcoming quantum device. Software development.                |  |                               |                                         |
| 2014-2015                                                                                        |  | Gymnasium Sorghvliet          | Physics teacher intern                  |
| Half-year teaching internship for high school physics. Culminated in official qualification.     |  |                               |                                         |

## Education and qualifications

|                                                                                                                   |  |               |                                    |                 |
|-------------------------------------------------------------------------------------------------------------------|--|---------------|------------------------------------|-----------------|
| 2020-2024                                                                                                         |  | TU Berlin, DE | PhD (Dr. Rer. Nat.)                | Summa cum laude |
| Thesis under Anna Pappa in her quantum communication & cryptography group.                                        |  |               |                                    |                 |
| 2016-2019                                                                                                         |  | TU Delft, NL  | MSc. Applied Physics               | GPA: 8.4        |
| Quantum technologies track, thesis under Barbara Terhal at QuTech.                                                |  |               |                                    |                 |
| 2014-2015                                                                                                         |  | TU Delft, NL  | High school teaching qualification | N/A             |
| High school physics teaching qualification with half-year internship at Gymnasium Sorghvliet during minor of BSc. |  |               |                                    |                 |
| 2012-2015                                                                                                         |  | TU Delft, NL  | BSc. Applied Physics               | GPA: 7.5        |
| Thesis under prof. M. Verhaegen at Delft Centre for Systems and Control.                                          |  |               |                                    |                 |

## Personal projects and extracurricular activities

|                                                                                                                                                      |                                                |
|------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------|
| Software projects                                                                                                                                    |                                                |
| Various software projects, e.g. the <a href="#">Graphstabilizer</a> package and <a href="#">Spotify2PlexPlaylistSyncer</a> .                         |                                                |
| Self-hosting                                                                                                                                         |                                                |
| Deploying, hosting & orchestrating <a href="#">my website</a> , multiuser <a href="#">Nextcloud</a> , Git-, media- and photo servers, and many more. |                                                |
| 2011-2025                                                                                                                                            | Tutor, teaching (assistant), designing courses |
| High school tutoring, being teaching assistant, giving retake courses, designing and giving MOOC's.                                                  |                                                |
| 2012-2017                                                                                                                                            | Various student committees                     |
| President of case tour- and symposium committee, member of various other committees.                                                                 |                                                |

## Links

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| <a href="#">Published</a> | <a href="#">arXiv</a> |
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| <a href="#">Published</a> | <a href="#">arXiv</a> |
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| <a href="#">Slides</a> | <a href="#">Gif</a> |
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## Publications

|             |                                     |                                             |
|-------------|-------------------------------------|---------------------------------------------|
| <b>2020</b> | <b><i>PRX Quantum</i> 1, 020325</b> | <b><i>F. Hahn, J. de Jong, A. Pappa</i></b> |
|-------------|-------------------------------------|---------------------------------------------|

*Anonymous Conference Key Agreement*

|             |                                      |                                                                    |
|-------------|--------------------------------------|--------------------------------------------------------------------|
| <b>2021</b> | <b><i>New J. Phys.</i> 23 083026</b> | <b><i>C. Thalacker, F. Hahn, J. de Jong, A. Pappa, S. Barz</i></b> |
|-------------|--------------------------------------|--------------------------------------------------------------------|

*Anonymous and secret communications in quantum networks.*

|             |                                     |                                                                                             |
|-------------|-------------------------------------|---------------------------------------------------------------------------------------------|
| <b>2022</b> | <b><i>PRX Quantum</i> 3, 040306</b> | <b><i>F. Grasselli, G. Murta, J. de Jong, F. Hahn, D. Bruß, H. Kampermann, A. Pappa</i></b> |
|-------------|-------------------------------------|---------------------------------------------------------------------------------------------|

*Secure Anonymous Conferencing in Quantum Networks*

|             |                               |                                                                |
|-------------|-------------------------------|----------------------------------------------------------------|
| <b>2023</b> | <b><i>Quantum</i> 7, 1117</b> | <b><i>J. de Jong, F. Hahn, J. Eiser, N. Walk, A. Pappa</i></b> |
|-------------|-------------------------------|----------------------------------------------------------------|

*Anonymous conference key agreement in linear quantum networks*

|             |                                             |                                                                           |
|-------------|---------------------------------------------|---------------------------------------------------------------------------|
| <b>2023</b> | <b><i>Phys. Rev. Research</i> 5, 033222</b> | <b><i>L. Rückle, J. Budde, J. de Jong, F. Hahn, A. Pappa, S. Barz</i></b> |
|-------------|---------------------------------------------|---------------------------------------------------------------------------|

*Experimental ACKA using linear cluster states*

|             |                                             |                                                                          |
|-------------|---------------------------------------------|--------------------------------------------------------------------------|
| <b>2024</b> | <b><i>Phys. Rev. Research</i> 6, 013330</b> | <b><i>J. de Jong, F. Hahn, N. Tcholtchev, M. Hauswirth, A. Pappa</i></b> |
|-------------|---------------------------------------------|--------------------------------------------------------------------------|

*Extracting GHZ states from linear cluster states*

|             |                                           |                                                   |
|-------------|-------------------------------------------|---------------------------------------------------|
| <b>2024</b> | (preprint) <b><i>arXiv:2410.03961</i></b> | <b><i>A. Burchardt, J. de Jong, L. Vandr </i></b> |
|-------------|-------------------------------------------|---------------------------------------------------|

*Algorithm to Verify Local Equivalence os Stabilizer States*

|             |                                        |                                                                                |
|-------------|----------------------------------------|--------------------------------------------------------------------------------|
| <b>2025</b> | <b><i>Phys. Rev. A</i> 111, 052449</b> | <b><i>L. Vandr , J. de Jong, F. Hahn, A. Burchardt, O. G hne, A. Pappa</i></b> |
|-------------|----------------------------------------|--------------------------------------------------------------------------------|

*Distinguishing graph states by the properties of their marginals.*

|             |                                             |                                                                                     |
|-------------|---------------------------------------------|-------------------------------------------------------------------------------------|
| <b>2025</b> | <b><i>Phys. Rev. Applied</i> 24, 054053</b> | <b><i>J. de Jong, S. Scheiner, N. Solomons, Z. Chaoui, D. Markham, A. Pappa</i></b> |
|-------------|---------------------------------------------|-------------------------------------------------------------------------------------|

*Anonymous and private parameter estimation in networks of quantum sensors*

## Selected talks

|             |                           |                                                                          |        |
|-------------|---------------------------|--------------------------------------------------------------------------|--------|
| <b>2021</b> | <b><i>PhD Seminar</i></b> | <b><i>Berlin School of Optical Sciences and Quantum Technologies</i></b> | Online |
|-------------|---------------------------|--------------------------------------------------------------------------|--------|

*Measurement-based and blind quantum computation*

|             |                              |                                                     |            |
|-------------|------------------------------|-----------------------------------------------------|------------|
| <b>2022</b> | <b><i>Research visit</i></b> | <b><i>Koch group (Freie Universit t Berlin)</i></b> | Berlin, DE |
|-------------|------------------------------|-----------------------------------------------------|------------|

*Graph states as quantum networks*

|             |                          |                                                                                 |              |
|-------------|--------------------------|---------------------------------------------------------------------------------|--------------|
| <b>2023</b> | <b><i>Conference</i></b> | <b><i>DPG spring meeting 2023, atomic physics &amp; quantum information</i></b> | Hannover, DE |
|-------------|--------------------------|---------------------------------------------------------------------------------|--------------|

*Extracting maximal entanglement from linear cluster states*

|             |                              |                                         |           |
|-------------|------------------------------|-----------------------------------------|-----------|
| <b>2023</b> | <b><i>Research visit</i></b> | <b><i>LIP6, Sorbonne Universit </i></b> | Paris, FR |
|-------------|------------------------------|-----------------------------------------|-----------|

*Extracting GHZ states from linear cluster states - to extract or not to extract*

|             |                       |                                                        |           |
|-------------|-----------------------|--------------------------------------------------------|-----------|
| <b>2023</b> | <b><i>Meeting</i></b> | <b><i>Yearly Quantum Internet Alliance meeting</i></b> | Paris, FR |
|-------------|-----------------------|--------------------------------------------------------|-----------|

*Anonymity in quantum network protocols*

|             |                               |                                                                                    |            |
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| <b>2024</b> | <b><i>Invited speaker</i></b> | <b><i>MATH+ Einstein workshop on Quantum Cryptography and Quantum Networks</i></b> | Berlin, DE |
|-------------|-------------------------------|------------------------------------------------------------------------------------|------------|

*Quantum Networking using graph theory*